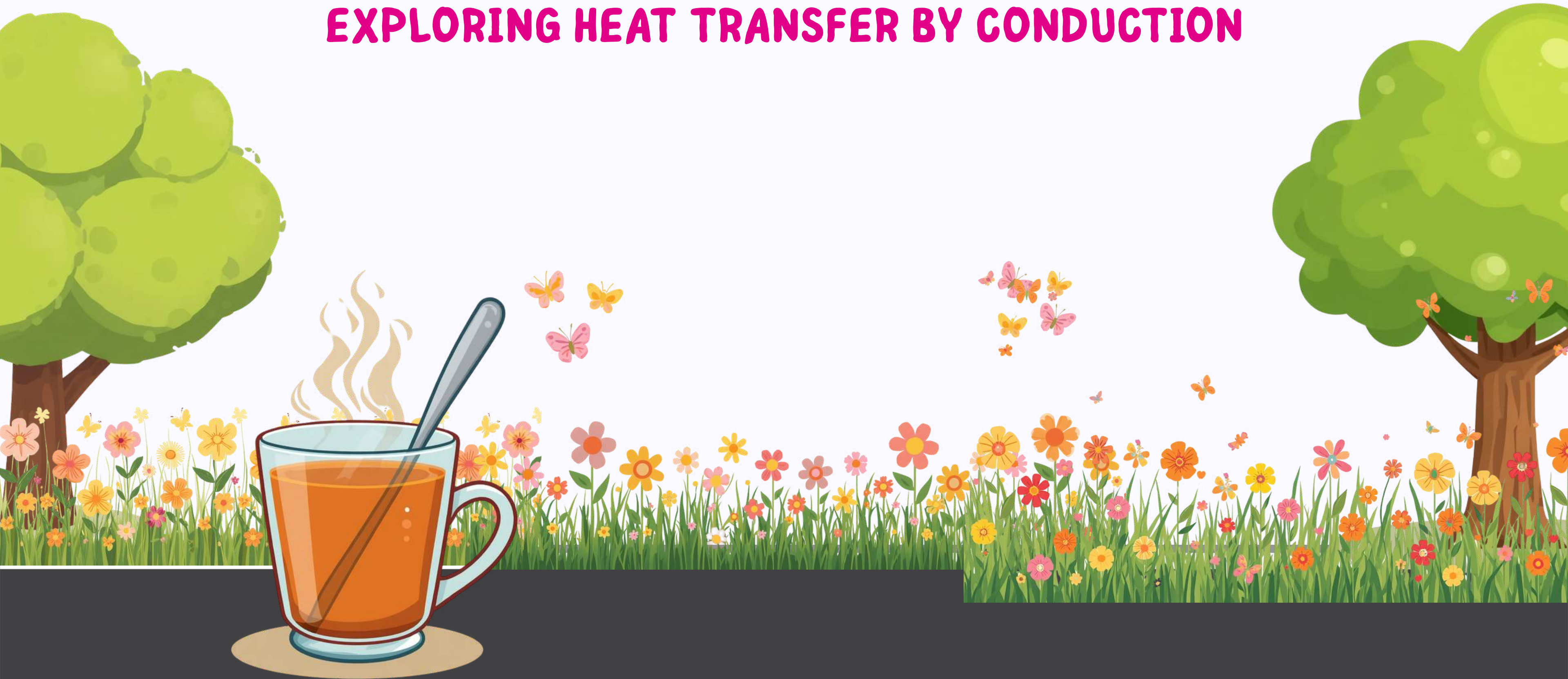


THE HOT SPOON MYSTERY

EXPLORING HEAT TRANSFER BY CONDUCTION

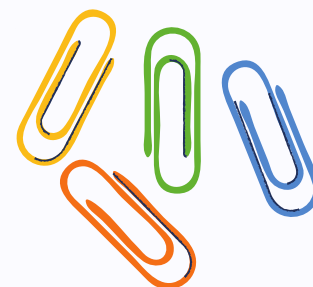




The Hot Spoon Mystery

Why does the metal spoon get hot when you put it in a cup of chai?

Can you guess how heat travels through the spoon? Think about it!



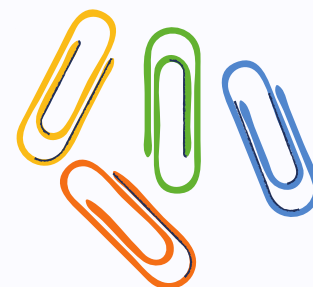


What is Conduction?

Conduction is heat traveling through things that touch each other.

Think of it like this: Heat is like tiny dancing particles passing energy hand-to-hand!

When something hot touches something cooler, the heat particles start dancing and pass their energy along, one to the next, like friends holding hands in a chain.





Conduction in Indian Kitchens

Heat travels through things that touch each other! Let's see how conduction happens in our kitchens every day when we cook delicious food.

- ✓ Mom making rotis on a hot Tawa — heat moves through the metal to cook the dough.
- ✓ Touching the metal ladle handle left in hot dal — it feels hot!





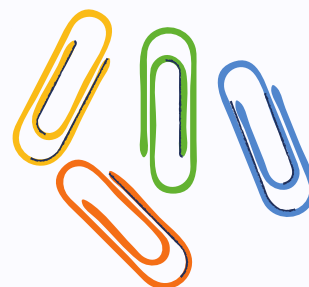
The Spoon Race

Demo Setup

Let's see conduction in action! We will put a metal spoon and a plastic spoon in hot water.

Think about it: Which spoon will get hot faster? Why do you think so?

Get ready to observe and discover!





What We Observe

The metal spoon gets hot quickly because metal is a conductor of heat. Heat travels fast through metal!

The plastic spoon stays cool because plastic is an insulator. It slows down heat like a speed bump!



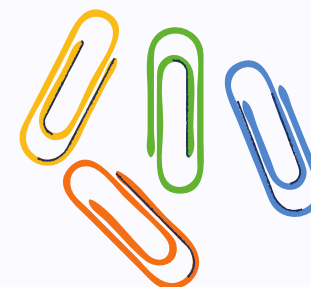


Conductors vs Insulators



Think of heat like a relay race!
Conductors pass heat quickly — like fast runners!
Insulators block heat — like a wall stopping the race!

- ✓ Fast Lanes (Conductors): Metal tawa, steel glass, iron kadai, aluminium foil
- ✓ Speed Bumps (Insulators): Wooden belan, plastic handle, rubber chappal, cotton cloth





Activity 1:

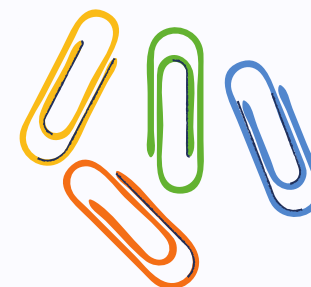
Kitchen Detective!

Can you find which of these items are conductors or insulators?

- 🔍 Steel glass (for drinking water)
- 🔍 Wooden rolling pin
- 🔍 Cloth napkin

Sort them into two groups with your classmates!

💡 Teacher tip: Discuss why each item belongs in its group.

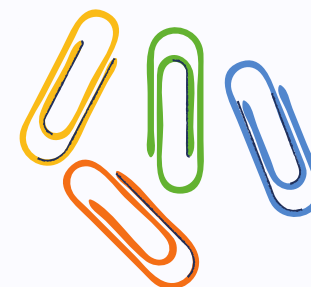




Touch or Don't Touch?

Let's think about safety in the kitchen! Some things are safe to touch when cooking, and some are not. Can you tell which is which? Remember: Conductors get hot fast, insulators stay cool!

- ✔ Is it safe to touch a metal ladle handle in hot dal? NO! Metal conducts heat fast.
- ✔ Can you touch the wooden belan handle when cooking? YES! Wood is an insulator.

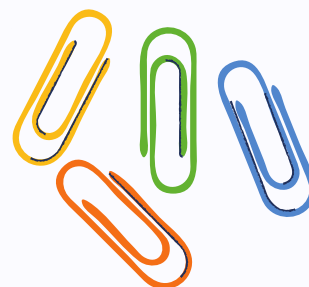




Safety First!

Always ask an adult before touching hot things in the kitchen. Never touch hot pots, pans, or spoons without help from a grown-up.

Remember: Science is fun but safety comes first! Be a smart explorer and stay safe!





How Heat Moves in Your Kitchen

Watch how heat travels by conduction!

Stove flame heats the metal Tawa

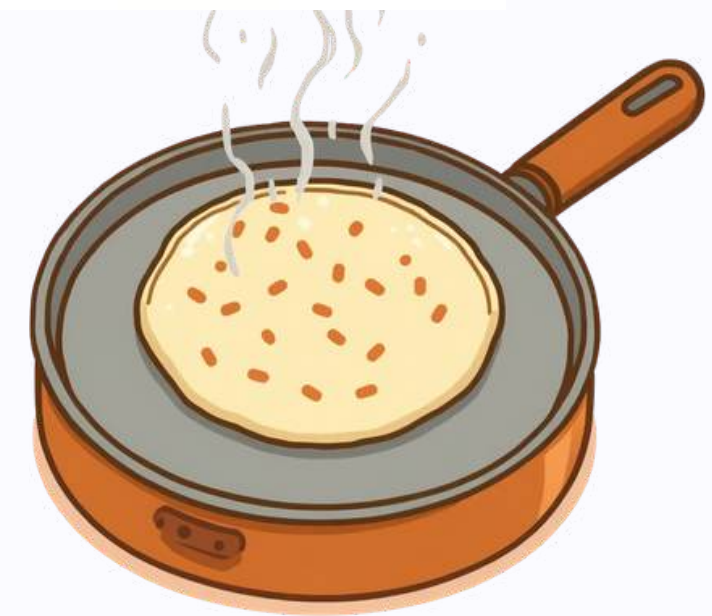
→ Heat moves through the Tawa

→ Tawa cooks the roti dough

Heat passes from one thing to the next
through touch - that's conduction in action!

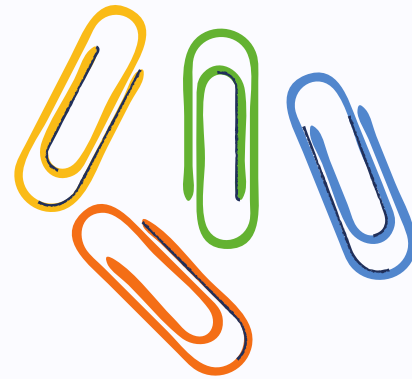


Heat flows by conduction



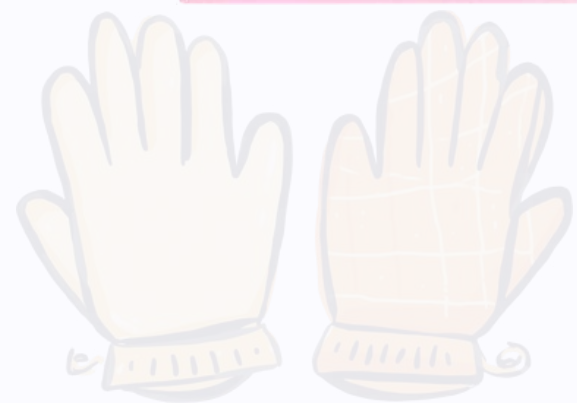


Summary Time!



Conduction

Heat moves through things that touch each other — this is called conduction!



Conductors

Metals like steel and iron let heat move fast through them!

Stay Safe!

Wood, plastic, and cloth slow heat down. Always ask an adult near hot things!





THANK YOU!

Any Questions About the Hot Spoon Mystery?

Try safe conduction observations at home with adult help!

Science Explorers!

